Brendan J. Clark, Ph.D.

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EDUCATION

Ph.D. Atmospheric Science January 2025
Rutgers University
Advisors: Dr. Alan Robock, Dr. Lili Xia
Dissertation: "Stratospheric Sulfate Aerosol Climate Intervention Implications for Global Agriculture"

M.S. Atmospheric Science November 2022
Rutgers University
Advisors: Dr. Alan Robock, Dr. Lili Xia
Thesis: "Optimal climate intervention scenarios for crop production vary by nation"

B.S. Environmental Science May 2020
University of Massachusetts
Minors: GIS, Natural Resource Conservation

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate April 2025-present
Cornell University

- Leveraging advanced statistical, machine learning, and process-based modeling to assess future impacts to tree growth
- Collaborating with CarbonPlan doing statistical downscaling of climate model simulations

Graduate Research Assistant August 2020-January 2025
Rutgers University

- Used the Community Earth System Model to assess impacts to global food production and nutritional quality under climate change
- Published results in *Nature Food, Earth's Future, Environmental Research Letters*
- Presented 15 first authors talks and posters

Undergraduate Research Assistant August 2018-May 2020
University of Massachusetts

- Used gas chromatography to analyze trace gas measurements from shale samples to determine greenhouse gas emissions from fracking
- Designed and administered a survey to assess how GIS is being taught by geocomputation instructors, publishing findings in *Research in Geographic Education*

PUBLICATIONS

PEER-REVIEWED ARTICLES (FIRST AUTHOR)

1. **B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Visioni, S. S. Rabin. Optimal climate intervention scenarios for crop production vary by nation. *Nature Food* 4, 902–911 **(2023)**. 10.1038/s43016-023-00853-3
2. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, G. Hoogenboom, J. Jaegermeyr. Maize yield changes under sulfate aerosol climate intervention using three global gridded crop models. *Earth's Future* 13, e2024EF005269 **(2025)**. 10.1029/2024EF005269
3. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, J. Jaegermeyr. Stratospheric aerosol climate intervention could reduce crop nutritional value. *Environmental Research Letters*, 20, 114083 **(2025)**. 10.1088/1748-9326/ae1151

PEER-REVIEWED ARTICLES (CO-AUTHOR)

1. F. Bowlick, **B. Clark**, et al. Understanding geocomputation education: A survey and syllabi informed review. Research in Geographic Education 23, 20-51 **(2022)**.
2. N. Grant, A. Robock, L. Xia, J. Singh, **B. Clark**. Impacts on Indian agriculture due to stratospheric aerosol intervention using agroclimatic indices. *Earth's Future* 13, e2024EF005262 **(2025)**. 10.1029/2024EF005262

FUNDING AND AWARDS

- Climate Intervention Network Travel Award ($1,000) 2025
- NSF NCAR Exploratory Allocation Award (500,000 CPU hours) 2025
- GeoMIP Annual Meeting Travel Funding Award ($3,500) 2025
- Rutgers Atmospheric Science Student Travel Support Award ($1,500) 2024
- Rutgers Atmospheric Science Student Travel Support Award ($1,200) 2023
- Rutgers Climate Institute Student Travel Support Award ($1,000) 2022
- NSF NCAR University Small Allocation Award (200,000 CPU hours) 2022
- Governor's Award in Recognition of Environmental Stewardship 2019
- John and Abigail Adams Scholarship Award ($7,000) 2016
- Stanley Z. Koplik Certificate of Mastery with Distinction Award 2016

TEACHING EXPERIENCE

Guest Lecturer Spring 2026
Cornell University
Graduate course EAS 5555, Theory and Practice of Earth System Modeling

Guest Lecturer Spring 2026
Rutgers University
Undergraduate honors seminar 01:090:297, Climate Intervention: Where Science Meets Ethics

Teaching Assistant Fall 2019
University of Massachusetts
NRC 585, Introduction to GIS

Substitute Teacher 2018-2018
Algonquin Regional High School

CONFERENCE PRESENTATIONS (FIRST AUTHOR)

1. **B Clark**, D. Visioni, S. Kothari, M. Lerdau. **Oral presentation**, March 2026. “Land models likely underestimate the effect of atmospheric dryness on European tree growth.” Coupled Model Intercomparison Project Community Workshop, Kyoto, Japan
2. **B Clark**, D. Visioni. **Poster**, March 2026. “Altered sunlight under stratospheric aerosol injection would have heterogenous impacts on vegetation.” Geoengineering Modeling Intercomparison Project annual meeting, Tokyo, Japan
3. **B Clark**, D. Visioni, S. Kothari, M. Lerdau. **Oral presentation**, December 2025. “Land models likely underestimate the effect of atmospheric dryness on European tree growth.” American Geophysical Union meeting, New Orleans, LA
4. **B Clark**, D. Visioni, S. Kothari, M. Lerdau. **Poster**, December 2025. “Using Dendrometer Data to Inform Projections of Carbon Storage under Solar adiation Modification.” American geophysical Union meeting, New Orleans, LA
5. **B. Clark. Poster**, May 2025 “Statistcally downscaled and bias adjusted ARISE-SAI and SSP2-4.5 scenarios.” DEGREES Global Forum, Cape Town, South Africa
6. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral presentation**, December 2024. “Stratospheric Aerosol Climate Intervention Could Reduce the Nutritional Value of Maize and Rice.” American Geophysical Union meeting, Washington D.C.
7. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Poster**, July 2024. “Sulfate Aerosol Climate Intervention Impacts on Maize Yield and Protein in Three Global Gridded Crop Models.” Geoengineering Modeling Intercomparison Project annual meeting, Cornell University
8. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral presentation**, February 2024. “Stratospheric Aerosol Climate Intervention Could Negatively Impact Crop Nutritional Quality.” Gordon Research Seminar, Newry, ME
9. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Poster**, February 2024. “Stratospheric Aerosol Climate Intervention Impacts on Crop Protein Content.” Gordon Research Conference, Newry, ME
10. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral Presentation**, December 2023. “Stratospheric Aerosol Climate Intervention Could Negatively Impact Crop Nutritional Quality.” American Geophysical Union meeting, San Francisco, CA
11. **B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Visioni, S. S. Rabin. **Oral presentation**, July 2023. “The Optimal Climate Intervention Scenario for Crop Production Varies by Nation.” Exteter, UK

12. **B. Clark,** A. Robock, L. Xia, S. Rabin, J. Singh, N. Grant. **Poster**, July 2023. “Rutgers Impact Studies of Climate Intervention (RISCI) Laboratory Group Overview.” Geoengineering Modeling Intercomparison Project annual meeting, Exeter, UK
13. **B. Clark,** A. Robock, L. Xia. **Oral presentation**, June 2023. “A Proposal for a Multi-Crop Model Assessment of Stratospheric Aerosol Climate Intervention.” AgMIP-GGCMI meeting, Columbia University
14. **B. Clark,** A. Robock, L. Xia. **Poster**, December 2022. “Can Crop Production be used as a Metric to Design Climate Intervention?” American Geophysical Union meeting, Chicago, IL
15. **B. Clark,** A. Robock, L. Xia. **Poster**, June 2022. “Impacts on Crop Production from Stratospheric Aerosol Climate Intervention: A Multi-Scenario Overview.” Gordon Research Conference, Newry, ME
16. **B. Clark,** A. Robock, L. Xia. **Oral presentation**, May 2022. “Crop Impacts from Stratospheric Aerosol Injection: A Multi-Scenario Overview.” University of Potsdam, Germany
17. **B. Clark,** A. Robock, L. Xia. **Oral presentation**, February 2022. “Discrepancies Between Fully Coupled and Offline CLM5 Crop Simulations.” NCAR Land Modeling Working Group Meeting, Boulder, CO
18. **B. Clark,** A. Robock, L. Xia. Poster, December 2021. “Impacts on Crop Production from Stratospheric Aerosol Injection” American Geophysical Union, New Orleans, LA
19. **B. Clark,** A. Robock, L. Xia. **Oral presentation**, October 2021. “Depicting Information and Remembering Your Audience: Impacts on Crop Production from Stratospheric Aerosol Climate Intervention” Climate Engineering in Context Conference, University of Potsdam, Germany

TECHNICAL SKILLS

- Python
    - Data analysis packages including xarray, numpy, pandas, zarr, scipy, and dask
    - Data visualization packages including matplotlib, seaborn, and plotly
    - Statistical modeling and machine learning packages including SciPy, statsmodels, scikit-learn, and seaborn
    - Jupyter notebooks and Google Colab
- Shell and command line – bash shell, batch scripting
- Parallelization using Python multiprocessing and Slurm scheduling
- High-performance computing environments including university clusters and NCAR’s Casper and Derecho
- Version control and collaborative work using Git and GitHub

SERVICE

- Journal refereeing for: Atmospheric Chemistry and Physics, Journal of Geophysical Research: Atmospheres, Environmental Research: Climate, PLOS Climate
- Reflective grant proposal reviewer
- Physicists Coalition for Nuclear Threat Reduction
- Science Policy and Advocacy at Rutgers

PROFESSIONAL AFFILIATIONS

- American Geophysical Union
- Ecological Society of America

PARTICIPATION IN INTERNATIONAL EXPERIMENTS

- Agricultural Modeling Intercomparison Project
- Global Gridded Crop Modeling Intercomparison Project
- Geoengineering Modeling Intercomparison Project
- Climate Intervention Biology Working Group
- Inter-Sectoral Impact Model Intercomparison Project